CSE305 Project: Parallel hierarchical clustering

Gleb Pogudin, gleb.pogudin@polytechnique.edu

Hierarchical clustering is a task in unsupervised machine learning aiming at organizing data into a tree-like data structures based on their similarities (languages into families, species into biological classification, items in a shop into categories, etc). The actual problem to be solved depends significantly on the notion of similarity (single-link, average-link, complete-link) and on the metric employed (one could cluster points in a high-dimensional space but also points with some specially-defined metric).

The goal of the project would be to take a hierarchical clustering problem (choosing the similarity for clusters and the metric employed), implement a sequential algorithm for solving it, and then design a parallel version of the algorithm (CPU or CUDA). As a starting point for possible ways to parallelize the most popular hierarchical clustering algorithms, we recommend the survey [2]. Note that this paper discusses algorithms in PRAM complexity model, so translating these algorithms into CPU setting would require thinking about the right split of the load. For further inspiration and ideas, we would recommend to more recent and more practice-oriented papers [1, 3]. Of course, you are welcome to do some more literature search.

The resulting algorithm has to be compared against the serial version for different number of threads on benchmarks of varying size. A collection of clustering datasets (but with labels which can be used for evaluating the result) is available at https://github.com/milaan9/Clustering-Datasets?tab=readme-ov-file, you are encouraged to find some datasets yourself.

For a group of three, either two different clustering problems should be considered or different algorithms should be tried for a single problem.

References

[1] M. Dash, S. Petrutiu, and P. Scheuermann. pPOP: Fast yet accurate parallel hierarchical clustering using partitioning. *Data & Knowledge Engineering*, 61(3):563–578, 2007. URL http://dx.doi.org/10.1016/j.datak.2006.07.004.

[2] C. F. Olson. Parallel algorithms for hierarchical clustering. *Parallel Computing*, 21(8): 1313–1325, Aug. 1995. ISSN 0167-8191. doi: 10.1016/0167-8191(95)00017-i. URL http://dx.doi.org/10.1016/0167-8191(95)00017-I.

[3] S. Rajasekaran. Efficient parallel hierarchical clustering algorithms. *IEEE Transactions on Parallel and Distributed Systems*, 16(6):497–502, 2005. URL https://doi.org/10.1109/TPDS.2005.72.